

MATH 200 - Elementary Linear Algebra

MATH 201 - Intermediate Linear Algebra

Section 7.2 Notes

Diagonalization

Devotional Thought

The Diagonalization Problem

The preceding section discussed the eigenvalue problem. In this section, we will look at another classic problem in linear algebra called the **diagonalization problem**; in terms of matrices, “for a square matrix A , does there exist an invertible matrix P such that $P^{-1}AP$ is diagonal?” Interestingly, the two questions are related!

Recall from Section 6.4 that two square matrices A and B are *similar* when there exists an invertible matrix P^{-1} such that $B = P^{-1}AP$.

Definition An $n \times n$ matrix A is **diagonalizable** when A is similar to a diagonal matrix. That is, A is diagonalizable when there exists an invertible matrix P such that $P^{-1}AP$ is a diagonal matrix.

With this definition, the diagonalization problem can be stated as

“Which square matrices are diagonalizable?”

Clearly, every diagonal matrix D is diagonalizable, because $D = I^{-1}DI$, where I is the identity matrix (of the same order as D).

As suggested in Example 8 in the preceding section, the eigenvalue problem is related closely to the diagonalization problem. The next two theorems shed more light on this relationship.

Theorem 7.4 If A and B are similar B matrices, then they have the same eigenvalues.

Proof. Suppose A and B are similar. Then there exists an invertible matrix P such that $B = P^{-1}BP$. Using various properties of matrices and determinants, it follows that

$$\begin{aligned} |\lambda I - B| &= |\lambda I - P^{-1}BP| \\ &= |\lambda(P^{-1}P) - P^{-1}BP| \\ &= |P^{-1}(\lambda I)P - P^{-1}BP| \\ &= |P^{-1}(\lambda I - A)P| \\ &= |P^{-1}| |\lambda I - A| |P| \\ &= |P^{-1}P| |\lambda I - A| \\ &= |I| |\lambda I - A| \\ &= |\lambda I - A|. \end{aligned}$$

That is, A and B have the same characteristic polynomial. Therefore, A and B have the same eigenvalues. ■

Theorem 7.4 does not, at first blush, seem all that useful...that is, until we recall that the eigenvalues of a triangular matrix are the main diagonal entries.

Example [Exercise 4] Given the matrices

$$A = \begin{bmatrix} 4 & -5 \\ 2 & -3 \end{bmatrix} \quad \text{and} \quad P = \begin{bmatrix} 1 & 5 \\ 1 & 2 \end{bmatrix},$$

(a) verify that A is diagonalizable by finding $P^{-1}AP$, and (b) use the result of part (a) and Theorem 7.4 to find the eigenvalues of A .

(a) Is $P^{-1}AP$ diagonal? $P^{-1} = \frac{1}{(1)(2)-(1)(5)} \begin{bmatrix} 2 & -5 \\ -1 & 1 \end{bmatrix}$

$$P^{-1}AP = \begin{bmatrix} -2/3 & 5/3 \\ 1/3 & -1/3 \end{bmatrix} \begin{bmatrix} 4 & -5 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} 1 & 5 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} -2/3 & 5/3 \\ 1/3 & -1/3 \end{bmatrix} \begin{bmatrix} 4 & -5 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} 1 & 5 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix} \checkmark$$

(b) A has the same eigenvalues as $P^{-1}AP$, which are $\lambda_1 = -1$ and $\lambda_2 = 2$.

Notice further: the column vectors of P are eigenvectors for A

$$A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4-5 \\ 2-3 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix} = \lambda_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$A \begin{bmatrix} 5 \\ 2 \end{bmatrix} = \begin{bmatrix} 20-10 \\ 10-6 \end{bmatrix} = \begin{bmatrix} 10 \\ 4 \end{bmatrix} = \lambda_2 \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

Theorem 7.5 Condition for Diagonalization

An $n \times n$ matrix A is diagonalizable if and only if it has n linearly independent eigenvectors.

The proof of this theorem is worth an examination as it tells exactly how to go about diagonalizing a square matrix.

Proof. ($\Rightarrow$) First, suppose A is diagonalizable. Then there exists an invertible matrix P such that $P^{-1}AP = D$ is diagonal. Letting the column vectors of P be $\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_n$, and the main diagonal entries of D be $\lambda_1, \lambda_2, \dots, \lambda_n$, produces

$$PD = \left[\begin{array}{c|c|c|c} & & & \\ \hline \mathbf{p}_1 & \mathbf{p}_2 & \cdots & \mathbf{p}_n \\ \hline & & & \end{array} \right] \left[\begin{array}{cccc} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{array} \right] = \left[\begin{array}{c|c|c|c} & & & \\ \hline \lambda_1 \mathbf{p}_1 & \lambda_2 \mathbf{p}_2 & \cdots & \lambda_n \mathbf{p}_n \\ \hline & & & \end{array} \right]$$

Now since $P^{-1}AP = D$ it follows that $AP = PD$, which implies

$$\left[\begin{array}{c|c|c|c} & & & \\ \hline A\mathbf{p}_1 & A\mathbf{p}_2 & \cdots & A\mathbf{p}_n \\ \hline & & & \end{array} \right] = \left[\begin{array}{c|c|c|c} & & & \\ \hline \lambda_1 \mathbf{p}_1 & \lambda_2 \mathbf{p}_2 & \cdots & \lambda_n \mathbf{p}_n \\ \hline & & & \end{array} \right].$$

That is, $A\mathbf{p}_i = \lambda_i \mathbf{p}_i$ for each column vector $\mathbf{p}_i$. Therefore the column vectors $\mathbf{p}_i$ of P are eigenvectors of A (and the diagonal entries λ_i are their associated eigenvalues). Moreover, P is invertible, so its column vectors are linearly independent. So, A has n linearly independent eigenvectors.

($\Leftarrow$) Conversely, assume A has n linearly independent eigenvectors $\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_n$ with corresponding eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$. Let P be the matrix whose columns are these n eigenvectors. That is,

$$P = \begin{bmatrix} | & | & \cdots & | \\ \mathbf{p}_1 & \mathbf{p}_2 & \cdots & \mathbf{p}_n \\ | & | & \cdots & | \end{bmatrix}.$$

Each $\mathbf{p}_i$ is an eigenvector of A , so we have $A\mathbf{p}_i = \lambda_i\mathbf{p}_i$ and thus

$$\begin{aligned} AP &= A \begin{bmatrix} | & | & \cdots & | \\ \mathbf{p}_1 & \mathbf{p}_2 & \cdots & \mathbf{p}_n \\ | & | & \cdots & | \end{bmatrix} \\ &= \begin{bmatrix} | & | & \cdots & | \\ \lambda_1\mathbf{p}_1 & \lambda_2\mathbf{p}_2 & \cdots & \lambda_n\mathbf{p}_n \\ | & | & \cdots & | \end{bmatrix} \\ &= \begin{bmatrix} | & | & \cdots & | \\ \mathbf{p}_1 & \mathbf{p}_2 & \cdots & \mathbf{p}_n \\ | & | & \cdots & | \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} \\ &= PD. \end{aligned}$$

Additionally, the column vectors for P , $\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_n$, are linearly independent (by assumption), so P is invertible. Thus the equation $AP = PD$ becomes $P^{-1}AP = D$, which means A is diagonalizable. ■

Example [Exercise 12] Find (if possible) a nonsingular matrix P such that $P^{-1}AP$ is diagonal. Verify that $P^{-1}AP$ is a diagonal matrix with the eigenvalues on the main diagonal.

$$A = \begin{bmatrix} 3 & 2 & -3 \\ -3 & -4 & 9 \\ -1 & -2 & 5 \end{bmatrix}$$

From last section:

$$\lambda_1 = 0 \quad E_{\lambda_1} = \text{span} \{ \underline{(-1, 3, 1)} \}$$

$$\lambda_2 = 2 \quad (\text{mult. } 2) \quad E_{\lambda_2} = \text{span} \{ \underline{(-3, 1, 0)}, \underline{(3, 0, 1)} \}$$

$$A = P D P^{-1}$$

with

$$D = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_2 \end{bmatrix}$$

$$P = \begin{bmatrix} -1 & -3 & 3 \\ 3 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 2 & -3 \\ -3 & -4 & 9 \\ 1 & -2 & 5 \end{bmatrix} = \begin{bmatrix} -1 & -2 & 3 \\ 3 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1/2 & 1 & -3/2 \\ -1/2 & -1 & 5/2 \\ -3/2 & -2 & 9/2 \end{bmatrix}$$

Steps for Diagonalizing a Square Matrix

Let A be an $n \times n$ matrix.

1. Find n linearly independent eigenvectors $\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_n$ for (if possible) with corresponding eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$. If n linearly independent eigenvectors do not exist, then A is not diagonalizable.

2. Let P be the $n \times n$ matrix whose columns consist of these eigenvectors. That is,

$$P = \begin{bmatrix} | & | & \cdots & | \\ \mathbf{p}_1 & \mathbf{p}_2 & \cdots & \mathbf{p}_n \\ | & | & \cdots & | \end{bmatrix}.$$

3. The diagonal matrix $D = P^{-1}AP$ will have the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ on its main diagonal. Note that the order of the eigenvectors used to form P will determine the order in which the eigenvalues appear on the main diagonal of D .

Example [Exercise 20] Show that the matrix is not diagonalizable.

$$\begin{bmatrix} 3 & 2 & -2 \\ 0 & -2 & 3 \\ 0 & 0 & -2 \end{bmatrix} \quad \lambda_1 = 3 \leftarrow \text{only gives a single eigenvector}$$

$\lambda_2 = -2$ (mult. 2) we have only 2 L.I. eigenvectors, not 3; not enough to diagonalize.

$$(\lambda_2 I - A) \bar{x}_2 = \bar{0} \Rightarrow \begin{bmatrix} 5 & -2 & 3 & | & 0 \\ 0 & 0 & -3 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 5 & -2 & 0 & | & 0 \\ 0 & 0 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} \Rightarrow \mathbf{v}_2 = \{2, -5, 0\}$$

only 1 vector!

“Dad why is my sisters name **Rose**?”

“Because your Mother loves roses”

“Thanks Dad”

“No problem PDP^{-1} ”



A Sufficient Condition for Diagonalization

Consider that the following conditions are equivalent for a square matrix A of order n :

- A is diagonalizable.
- A has n linearly independent eigenvectors.
- The sum of the dimensions of the eigenspaces must be equal to n .

The latter condition requires justification. What is implicit to our understanding of the eigenvalue problem is the following claim:

Claim If $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ are eigenvectors of A having *distinct* eigenvalues, then the eigenvectors must be linearly independent.

Put another way: the eigenspaces associated with different eigenvalues cannot overlap (except at the zero vector). This means that there is no risk of over counting basis vectors for the various eigenspaces, meaning we can count them up to see if there are enough.

The above claim is verified in the proof of the following theorem; the statement goes even further in that it gives a *sufficient* (but not *necessary*) condition for diagonalization of a matrix.

That is, if $\mu_A(\lambda) = 1$ for all λ , then $\gamma_A(\lambda) = 1$ for all λ , and thus there are enough e-vectors to diagonalize.

Theorem 7.6 Sufficient Condition for Diagonalization

If an $n \times n$ matrix A has n *distinct* eigenvalues, then the corresponding eigenvectors are linearly independent and A is diagonalizable.

In general, A is diagonalizable if and only if $\mu_A(\lambda) = \gamma_A(\lambda)$ for all eigenvalues λ of A .

Proof of Theorem 7.6. Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be n distinct eigenvalues of A with corresponding eigenvectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$. Since we wish to show that these vectors are linearly independent, suppose that

$$c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \dots + c_n\mathbf{x}_n = \mathbf{0}. \quad (1)$$

Our goal is to show that $c_1 = c_2 = \dots = c_n = 0$.

Consider multiplying Equation (1) by $(\lambda_1 I - A)$. Since each vector $\mathbf{x}_i$ has an associated eigenvalue λ_i it follows that $A\mathbf{x}_i = \lambda_i\mathbf{x}_i$ for each $i = 1, 2, \dots, n$. Thus the first term in Equation (1) will become zero:

$$(\lambda_1 I - A)(c_1\mathbf{x}_1) = c_1(\lambda_1\mathbf{x}_1 - A\mathbf{x}_1) = c_1(\lambda_1\mathbf{x}_1 - \lambda_1\mathbf{x}_1) = \mathbf{0}$$

In a similar manner, Equation (1) thus becomes

$$c_2(\lambda_1 I - \lambda_2)\mathbf{x}_2 + \dots + c_n(\lambda_1 I - \lambda_n)\mathbf{x}_n = \mathbf{0}. \quad (2)$$

This process can be repeated to eliminate each term in the sum until we are left with only the final term: We would next multiply Equation (2) by $(\lambda_2 I - A)$ which would force the first term to be zero, thereby leaving a new equation (Equation 3, I suppose) with terms of the form

$$c_i(\lambda_1 I - \lambda_i)(\lambda_2 I - \lambda_i)\mathbf{x}_i.$$

Continuing in this way leaves only one final term equal to zero (Equation n):

$$c_n(\lambda_1 - \lambda_n)(\lambda_2 - \lambda_n) \dots (\lambda_{n-1} - \lambda_n) = 0$$

Now we finally use our assumption that the eigenvalues are distinct to conclude that $c_n = 0$.

Starting back from Equation (1), we can use the above process to eliminate *any* of the terms one-by-one leaving only a term involving c_k (for some $k = 1, 2, \dots, n$) and a bunch of nonzero factors whose product equals zero. Hence, it can be shown that $c_k = 0$ for *all* $k = 1, 2, \dots, n$. Therefore, the vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ must be linearly independent.

Moreover, since the matrix A has n linearly independent eigenvectors, Theorem 7.5 guarantees that A is diagonalizable. ■

Remark The condition in Theorem 7.6 is sufficient but not necessary for diagonalization. In other words, a diagonalizable matrix need not have distinct eigenvalues. We saw examples of matrices with non-distinct eigenvalues (multiplicity greater than 1) in Exercises 12 and 20; the former was diagonalizable while the latter was not.

Examples Find the eigenvalues of the matrix and determine whether there is a sufficient number of eigenvalues to guarantee that the matrix is diagonalizable by Theorem 7.6.

[Exercise 24] $\begin{bmatrix} 2 & 0 \\ 5 & 2 \end{bmatrix}$ only 1 eigenvalue: $\lambda = 2$

This is not enough to say whether the matrix is diagonalizable or not. However,

$$(\lambda I - A)\vec{x} \Rightarrow \begin{bmatrix} 0 & 0 & | & 0 \\ 5 & 0 & | & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix} \Rightarrow B = \{ \underline{(0, 1)} \}$$

[Example 7] $\begin{bmatrix} 1 & -2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & -3 \end{bmatrix}$ $\lambda = 1, 0, -3$

Not enough L.I. eigenvectors to diagonalize.

3 eigenvectors for an order 3 matrix

The matrix must be diagonalizable.

Assuming $T: V \rightarrow V$ is diagonalizable ...

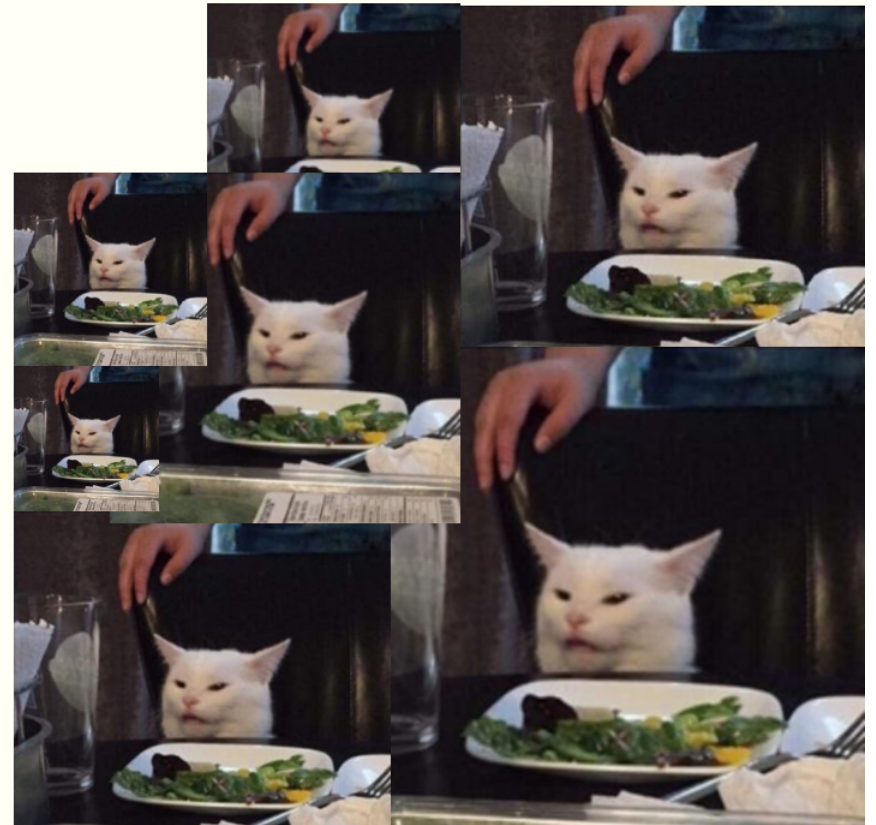
A specific linear transformation matrix exists

Other vectors



T in most bases for V

eigen vectors of that matrix



T in the eigenbasis for V (here note that all λ would be positive)

Diagonalization of Linear Transformations

So far in this section, the diagonalization problem has been in terms of matrices. In terms of linear transformations, the diagonalization problem can be stated as: For a linear transformation $T : V \rightarrow V$ does there exist a basis B for V such that the matrix for T relative to B is diagonal? The answer is "yes" when the standard matrix for T is diagonalizable.

Examples [Example 8] Let $T : R^3 \rightarrow R^3$ be the linear transformation represented by

$$T(x_1, x_2, x_3) = (x_1 - x_2 - x_3, x_1 + 3x_2 + x_3, -3x_1 + x_2 - x_3).$$

If possible, find a basis B for R^3 such that the matrix for T relative to B is diagonal.

we need some matrix A for T with respect to some basis; use the standard matrix:

$$T(1, 0, 0) = (1, 1, -3)$$

$$T(0, 1, 0) = (-1, 3, 1)$$

$$T(0, 0, 1) = (-1, 1, -1)$$

$$\Rightarrow A = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 3 & 1 \\ -3 & 1 & -1 \end{bmatrix}$$

The basis for T which diagonalizes T is

Solving the eigen value problem:

$$\lambda_1 = 2 \quad B_1 = \{(-1, 0, 1)\}$$

$$\lambda_2 = -2 \quad B_2 = \{(1, -1, 4)\}$$

$$\lambda_3 = 3 \quad B_3 = \{(-1, 1, 1)\}$$

$$B = \{(-1, 0, 1), (1, -1, 4), (-1, 1, 1)\}$$

The matrix for T relative to B is

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$